VTPEH 6270 - Check Point 02

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Contents

```r
library(tidyverse)
library(readr)

brfss <- readr::read_csv("NYSDOH_BRFSS_SurveyData_2023.csv", show_col_types = FALSE)
dim(brfss)
```

```
## [1] 17349   370
```

```r
vars_keep <- c("AGE", "SEXVAR", "BPHIGH6", "BPMEDS1", "_BMI5CAT", "_BMI5", "_LLCPWT")


df <- brfss %>%
  dplyr::select(dplyr::all_of(vars_keep))

glimpse(df)
```

```
## Rows: 17,349
## Columns: 7
## $ AGE        <chr> "Age 45 - 54", "Age 65 or older", "Age 65 or older", "Age 6~
## $ SEXVAR     <chr> "Male", "Female", "Male", "Female", "Female", "Female", "Fe~
## $ BPHIGH6    <chr> "No", "Yes", "Yes", "Yes", "No", "No", "Told borderline hig~
## $ BPMEDS1    <chr> "Not asked or Missing", "Yes", "Yes", "Yes", "Not asked or ~
## $ `_BMI5CAT` <chr> "Obese", "Normal Weight", "Normal Weight", "Don\x92t know/R~
## $ `_BMI5`    <chr> "1 or greater", "1 or greater", "1 or greater", "Don\x92t k~
## $ `_LLCPWT`  <dbl> 3425.1990, 975.3992, 1184.7724, 182.4274, 964.8836, 11831.8~
```

```r
df_clean <- df %>%
  dplyr::rename(
    age_group = AGE,
    sex = SEXVAR,
    bp_high = BPHIGH6,
    bp_meds = BPMEDS1,
    bmi_cat = `_BMI5CAT`,
    weight = `_LLCPWT`
  ) %>%
  dplyr::filter(
    !stringr::str_detect(age_group, "Refused|Missing|Not sure|Don't know"),
```

```r
    !stringr::str_detect(sex, "Refused|Missing|Not sure|Don't know"),
    !stringr::str_detect(bp_high, "Refused|Missing|Not sure|Don't know|Not asked"),
    !stringr::str_detect(bp_meds, "Refused|Missing|Not sure|Don't know|Not asked"),
    !stringr::str_detect(bmi_cat, "Refused|Missing|Not sure|Don't know")
  )

glimpse(df_clean)
```

```
## Rows: 5,424
## Columns: 7
## $ age_group <chr> "Age 65 or older", "Age 65 or older", "Age 65 or older", "Ag~
## $ sex       <chr> "Female", "Male", "Male", "Female", "Male", "Male", "Male", ~
## $ bp_high   <chr> "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes~
## $ bp_meds   <chr> "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes~
## $ bmi_cat   <chr> "Normal Weight", "Normal Weight", "Overweight", "Obese", "Ov~
## $ `_BMI5`   <chr> "1 or greater", "1 or greater", "1 or greater", "1 or greate~
## $ weight    <dbl> 975.39917, 1184.77242, 472.56019, 7677.91113, 253.68825, 249~
```

```r
variable_overview <- tibble(
  Variable = names(df_clean),
  Type = if_else(sapply(df_clean, is.numeric), "Continuous", "Categorical"),
  Class = sapply(df_clean, function(x) paste(class(x), collapse = ", "))
)

variable_overview
```

```
## # A tibble: 7 x 3
##   Variable  Type        Class
##   <chr>     <chr>       <chr>
## 1 age_group Categorical character
## 2 sex       Categorical character
## 3 bp_high   Categorical character
## 4 bp_meds   Categorical character
## 5 bmi_cat   Categorical character
## 6 _BMI5     Categorical character
## 7 weight    Continuous  numeric
```

```r
df_clean %>%
  count(bmi_cat, sort = TRUE)
```

```
## # A tibble: 4 x 2
##   bmi_cat           n
##   <chr>         <int>
## 1 Obese          2225
## 2 Overweight     2000
## 3 Normal Weight  1133
## 4 Underweight      66
```

```r
df_clean %>%
  count(bp_high, sort = TRUE)
```

```
## # A tibble: 1 x 2
##   bp_high     n
##   <chr>   <int>
## 1 Yes      5424
```

```
df_clean %>%
  count(bp_meds, sort = TRUE)
```

```
## # A tibble: 3 x 2
##   bp_meds                      n
##   <chr>                    <int>
## 1 "Yes"                     4319
## 2 "No"                      1096
## 3 "Don\x92t know/Not Sure"     9
```

Most individuals with high blood pressure reported taking blood pressure medication, although a substantial portion were not currently using medication.

```
df_clean %>%
  count(sex, sort = TRUE)
```

```
## # A tibble: 2 x 2
##   sex        n
##   <chr>  <int>
## 1 Male    2946
## 2 Female  2478
```

```
df_clean %>% count(age_group, sort = TRUE)
```

```
## # A tibble: 6 x 2
##   age_group           n
##   <chr>           <int>
## 1 Age 65 or older  2702
## 2 Age 55 - 64      1247
## 3 Age 45 - 54       695
## 4 Age 35 - 44       421
## 5 Age 25 - 34       254
## 6 Age 18 - 24       105
```

```
df_clean %>%
  count(`_BMI5`, sort = TRUE)
```

```
## # A tibble: 1 x 2
##   `_BMI5`          n
##   <chr>        <int>
## 1 1 or greater  5424
```

```
summary(df_clean$weight)
```

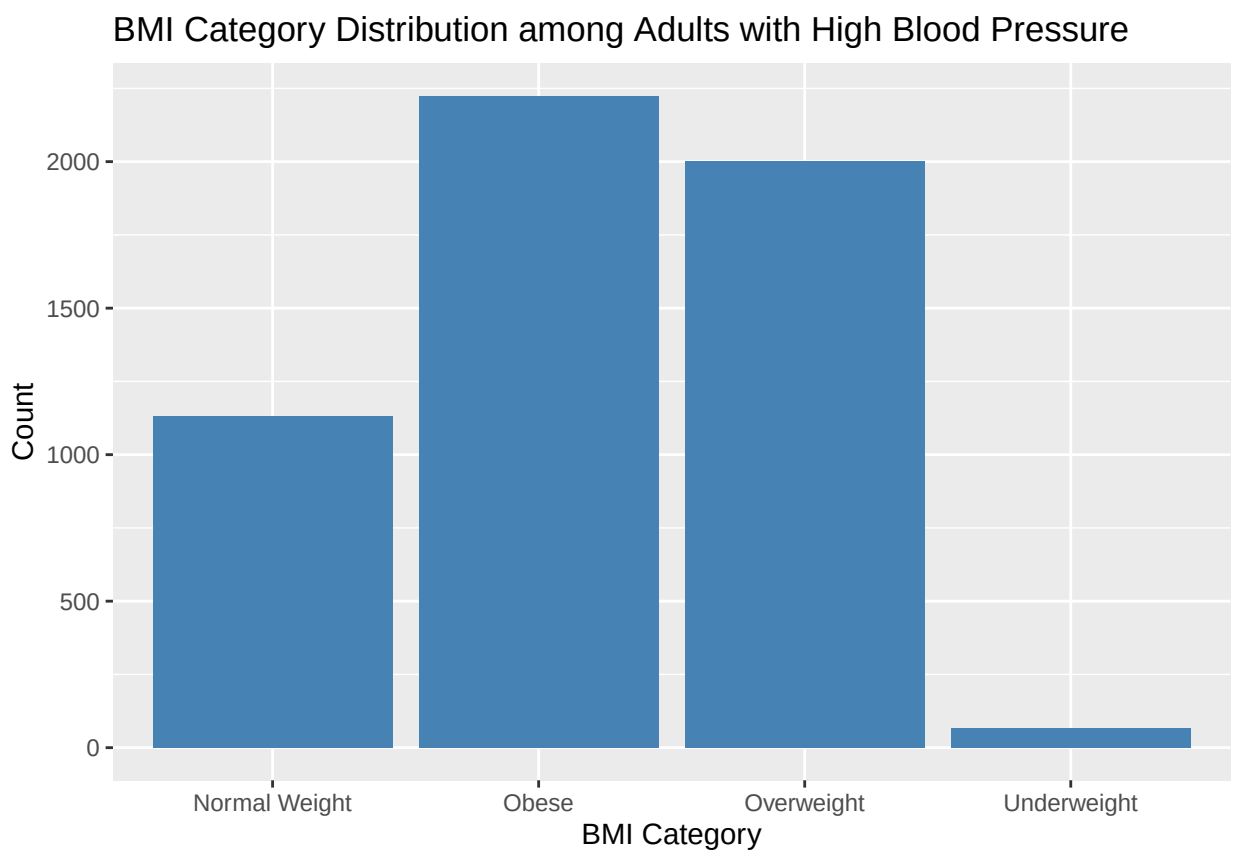
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    11.6   252.4   515.2   808.7   960.8 17896.4
```

Body weight showed a right-skewed distribution, with a mean weight higher than the median, indicating the presence of some extremely high values in the dataset.

```
df_clean %>% count(bmi_cat, sort = TRUE)
```

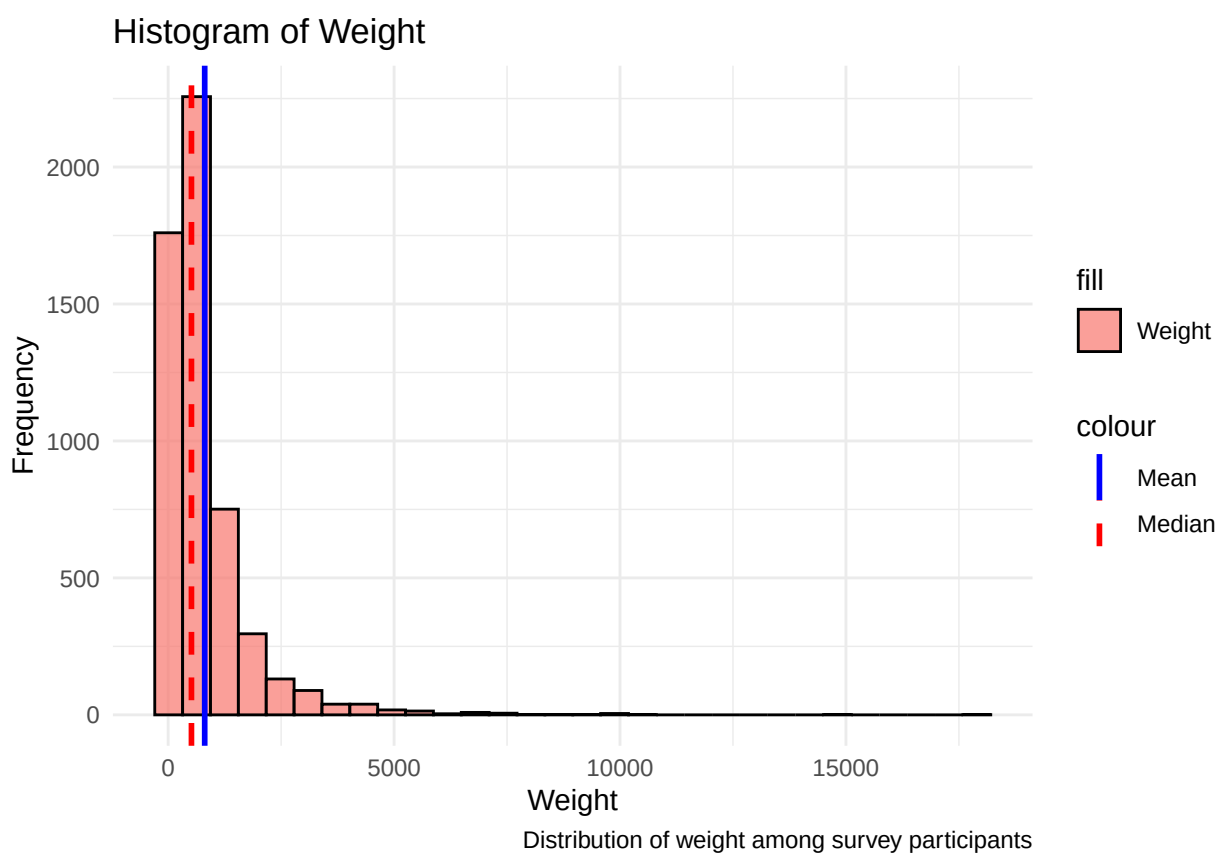
```
## # A tibble: 4 x 2
##   bmi_cat           n
##   <chr>         <int>
## 1 Obese          2225
## 2 Overweight     2000
## 3 Normal Weight  1133
## 4 Underweight      66
```

```
ggplot(df_clean, aes(x = bmi_cat)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "BMI Category Distribution among Adults with High Blood Pressure",
    x = "BMI Category",
    y = "Count"
  )
```



The majority of adults with high blood pressure were classified as either obese or overweight. Obese individuals represented the largest group, followed closely by those who were overweight. Normal weight individuals were less common, while underweight individuals made up a very small proportion of the sample. This suggests a strong association between elevated body weight and high blood pressure in this population.

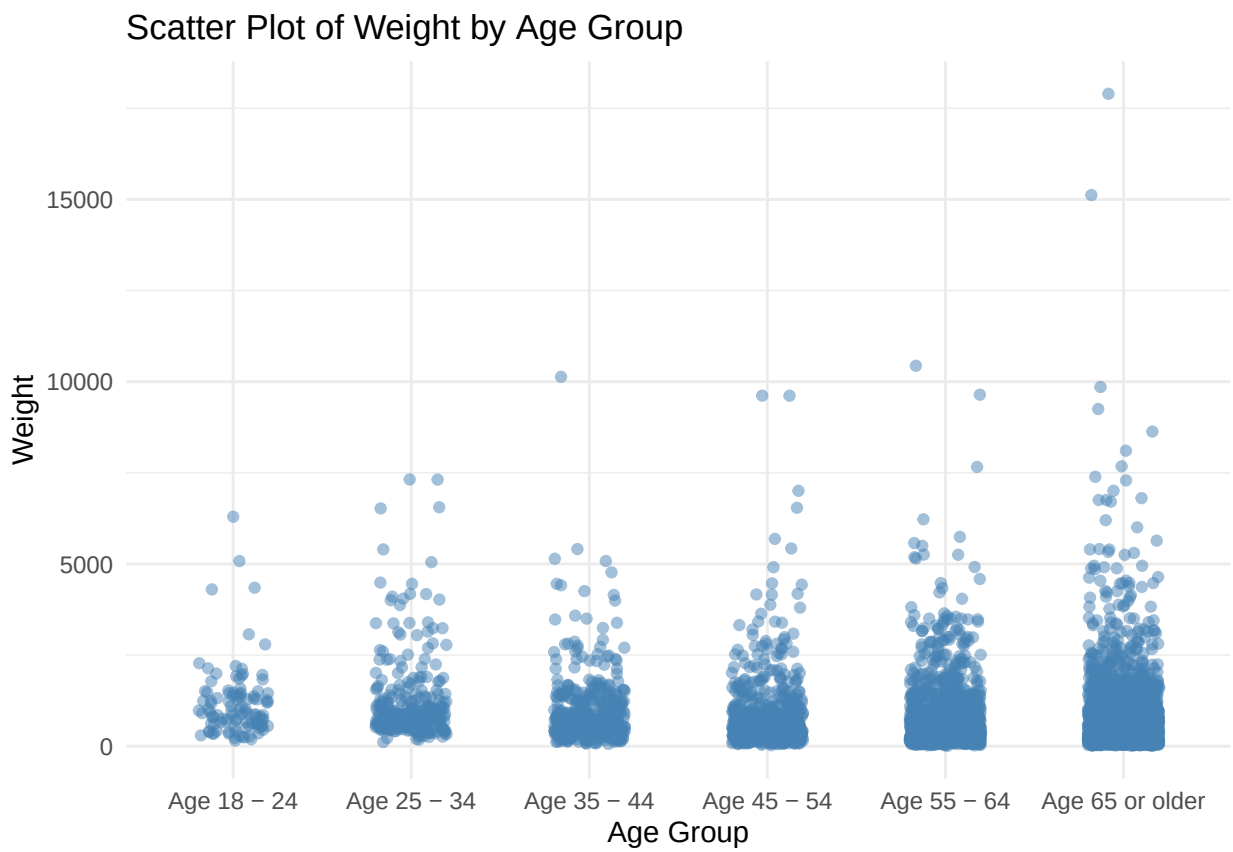
```r
library(ggplot2)

mean_w <- mean(df_clean$weight, na.rm = TRUE)
median_w <- median(df_clean$weight, na.rm = TRUE)

ggplot(df_clean, aes(x = weight)) +
  geom_histogram(aes(fill = "Weight"), bins = 30, color = "black", alpha = 0.7) +
  geom_vline(aes(xintercept = mean_w, color = "Mean"), linewidth = 1) +
  geom_vline(aes(xintercept = median_w, color = "Median"), linewidth = 1, linetype = "dashed") +
  scale_color_manual(values = c("Mean" = "blue", "Median" = "red")) +
  labs(
    x = "Weight",
    y = "Frequency",
    title = "Histogram of Weight",
    caption = "Distribution of weight among survey participants"
  ) +
  theme_minimal()
```



The histogram shows the distribution of weight among survey participants with high blood pressure. Most observations are concentrated at lower weight values, with a long right tail indicating a right-skewed distribution. The mean weight is higher than the median, suggesting the presence of extreme high-weight outliers. This implies that while most individuals fall within a moderate weight range, a small number of participants have very high weights.

```r
ggplot(df_clean, aes(x = age_group, y = weight)) +
  geom_jitter(alpha = 0.5, width = 0.2, color = "steelblue") +
```

```r
labs(
  title = "Scatter Plot of Weight by Age Group",
  x = "Age Group",
  y = "Weight"
) +
theme_minimal()
```



The scatter plot illustrates the relationship between weight and age group among individuals with high blood pressure. Each point represents an individual's weight within a specific age category. The spread of weight values increases in older age groups, particularly among those aged 55 and above. This suggests greater variability in body weight among older adults, with several high-weight outliers appearing in the older age categories.

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